Recitation (1)

Problem 1.1:

Express the unit vector which is directed toward the origin from an arbitrary point on the plane $z = -5$

Express the unit vector which points from $z = h$ on the z-axis toward$(\rho, \phi, 0)$in cylindrical coordinates.

Problem 1.2:

A circle, centered at the origin with a radius of 2 units, lies in the xy plane. Determine the unit vector in rectangular components that lies in the xy plane, is tangent to the circle at $(-\sqrt{3}, 1, 0)$, and is in the general direction of increasing values of y.

Problem 1.3:

Three vectors extending from the origin are given as $\overrightarrow{r_1} = (7, 3, -2)$, $\overrightarrow{r_2} = (-2, 7, -3)$, and $\overrightarrow{r_3} = (0, 2, 3)$. Find

a) a unit vector perpendicular to both $\overrightarrow{r_1}$ and $\overrightarrow{r_2}$
b) the area of the triangle defined by $\overrightarrow{r_1}$ and $\overrightarrow{r_2}$

Problem 1.4:

Given that $\vec{A} = 3\hat{\rho} + 2\hat{\phi} + \hat{z}$ and $\vec{B} = 5\hat{\rho} - 8\hat{z}$, find: (a) $\vec{A} + \vec{B}$, (b) $\vec{A} \cdot \vec{B}$, (c) $\vec{A} \times \vec{B}$,(d) the angle between $\vec{A}$ and $\vec{B}$, (e) the area of the parallelogram defined by $\vec{A}$ and $\vec{B}$

Problem 1.5:

Given $\vec{A} = \hat{x} + \hat{y}$, $\vec{B} = \hat{x} + 2\hat{z}$, and $\vec{C} = 2\hat{y} + \hat{z}$,

a) Find $\vec{A} \times (\vec{B} \times \vec{C})$ and compare it with $(\vec{A} \times \vec{B}) \times \vec{C}$
b) Find $\vec{A}.(\vec{B} \times \vec{C})$ and compare it with $\vec{B}.(\vec{C} \times \vec{A})$, deduce the volume of the parallelepiped defined by $\vec{A}$, $\vec{B}$ and $\vec{C}$

Problem 1.6:

If $\vec{J} = r\sin\theta\cos\phi\hat{r} - \cos 2\theta\sin\phi\hat{\theta} + \tan\frac{\theta}{2}\ln r\hat{\phi}$ at $T(2, \pi/2, 3\pi/2)$, determine the vector component of $\vec{J}$ that is:

a) Parallel to $\hat{z}$
b) Normal to surface $\phi = \frac{3\pi}{2}$
c) Tangential to the spherical surface $r = 2$
d) Parallel to the line $y = -2, z = 0$

If $\vec{H} = 3\rho\hat{\rho} - \rho cos\phi\hat{\phi} - z^2\hat{z}$, find at $(1, 60°, 2)$ the component of $\vec{H}$ parallel to surface $\rho = 1$

Problem 1.7:

Express the following vectors in rectangular coordinates:

a) $\vec{A} = \rho\sin\phi\hat{\rho} + \rho\cos\phi\hat{\phi} - 2z\hat{z}$
b) $\vec{B} = 4r\cos\phi\hat{r} + r\hat{\theta}$

Problem 1.8:

Transform the following vectors to cylindrical coordinates:

a) $\vec{B} = \mathrm{r}\sin\theta\hat{r} - r^2\cos\phi\hat{\phi}$

b) $\vec{H} = \sqrt{x^2+y^2}\hat{x} + \frac{y}{\sqrt{x^2+y^2}}\hat{y} + z\hat{z}$

Problem 1.9:

Use the spherical coordinate system to find the area of the strip $\alpha \leq \theta \leq \beta$ on the spherical shell of radius a. What results when $\alpha = 0$ and $\beta = \pi$?

Problem 1.10:

The surfaces $\rho = 3, \rho = 5, \phi = 100°, \phi = 130°, z = 3$, and $z = 4.5$ define a closed surface. Find

a) the enclosed volume;
b) the total area of the enclosing surface;
c) the total length of the twelve edges of the surfaces;

Problem 1.11:

The surfaces $r = 2$ and $4, \theta = 30°$ and $50°$, and $\phi = 20°$ and $60°$ identify a closed surface. Find

a) the enclosed volume;
b) the total area of the enclosing surface;
c) the total length of the twelve edges of the surface;